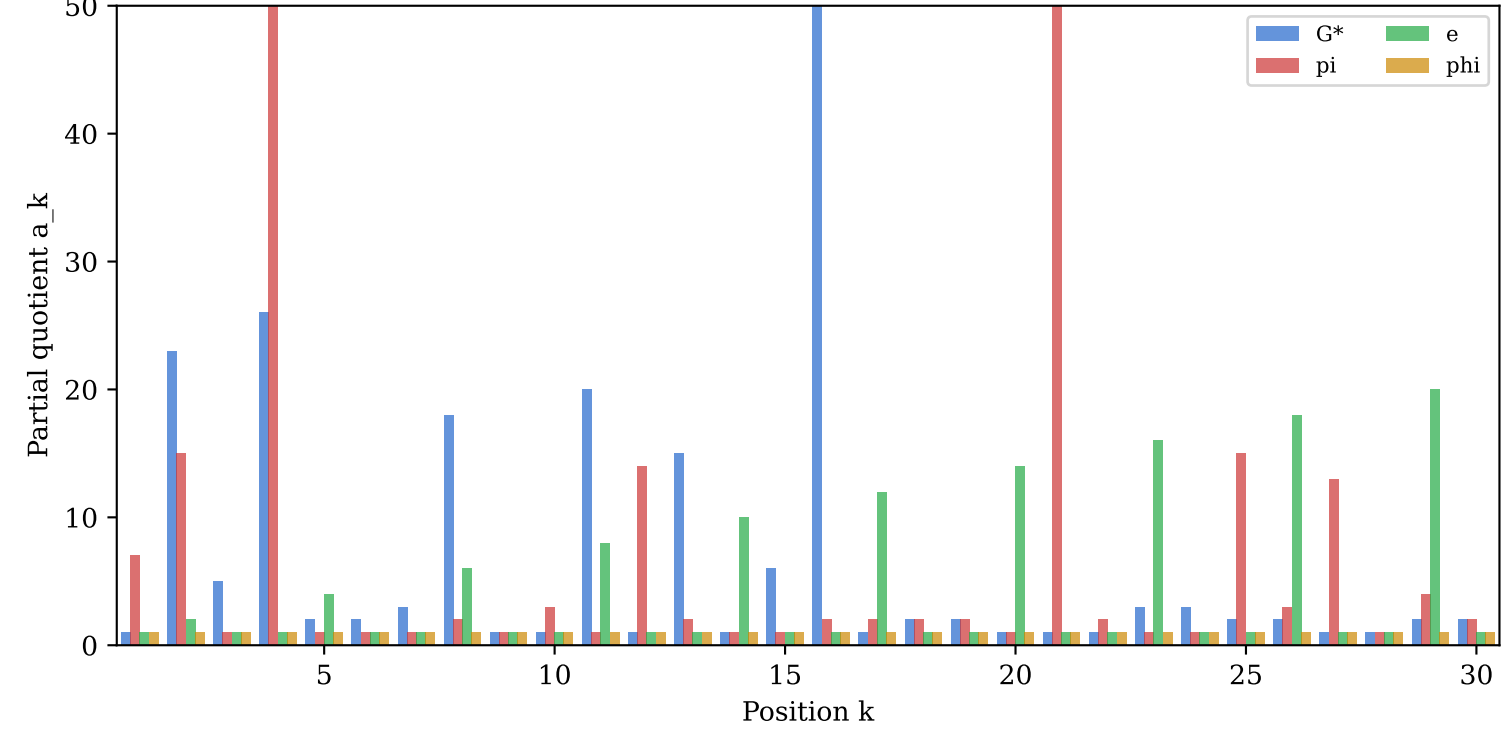
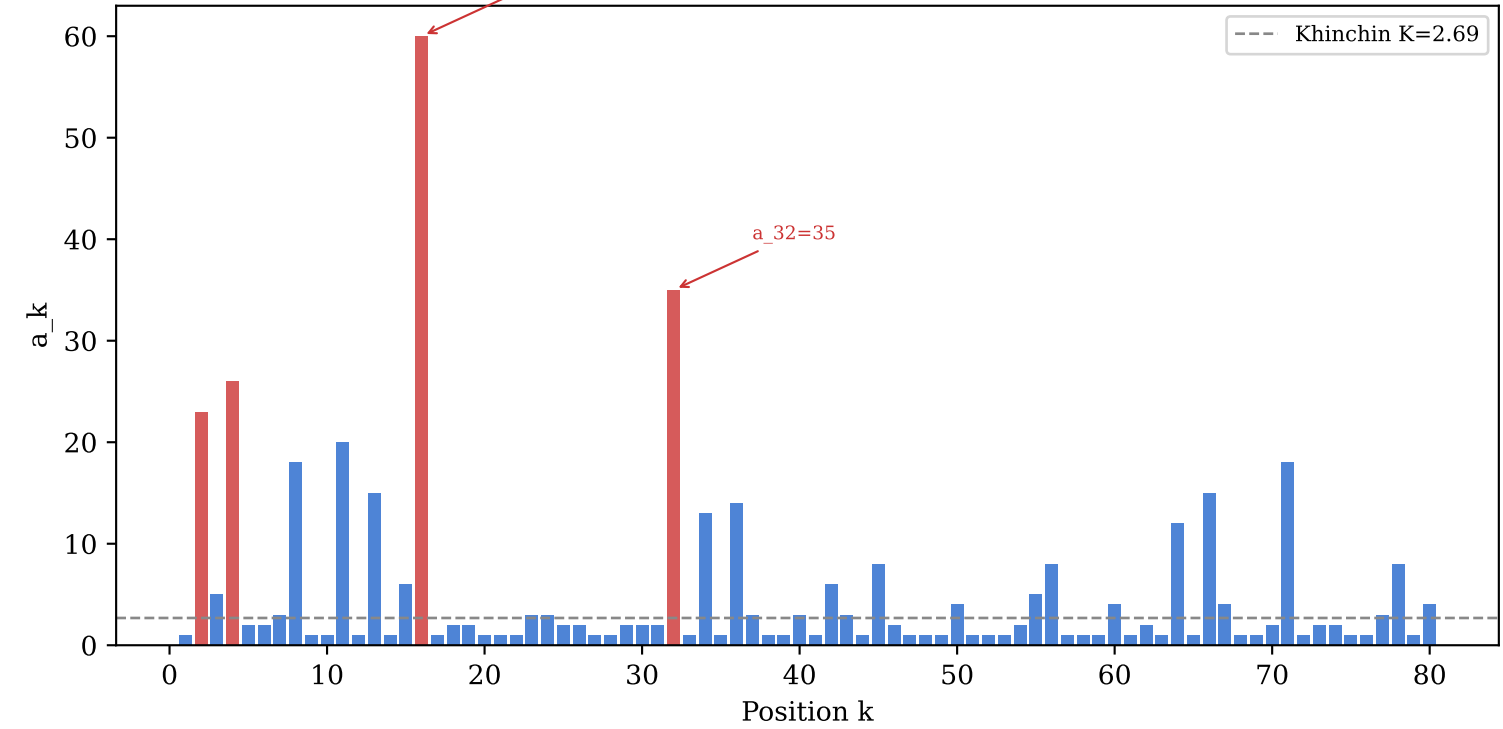


Continued Fraction Structure of G* and Friends

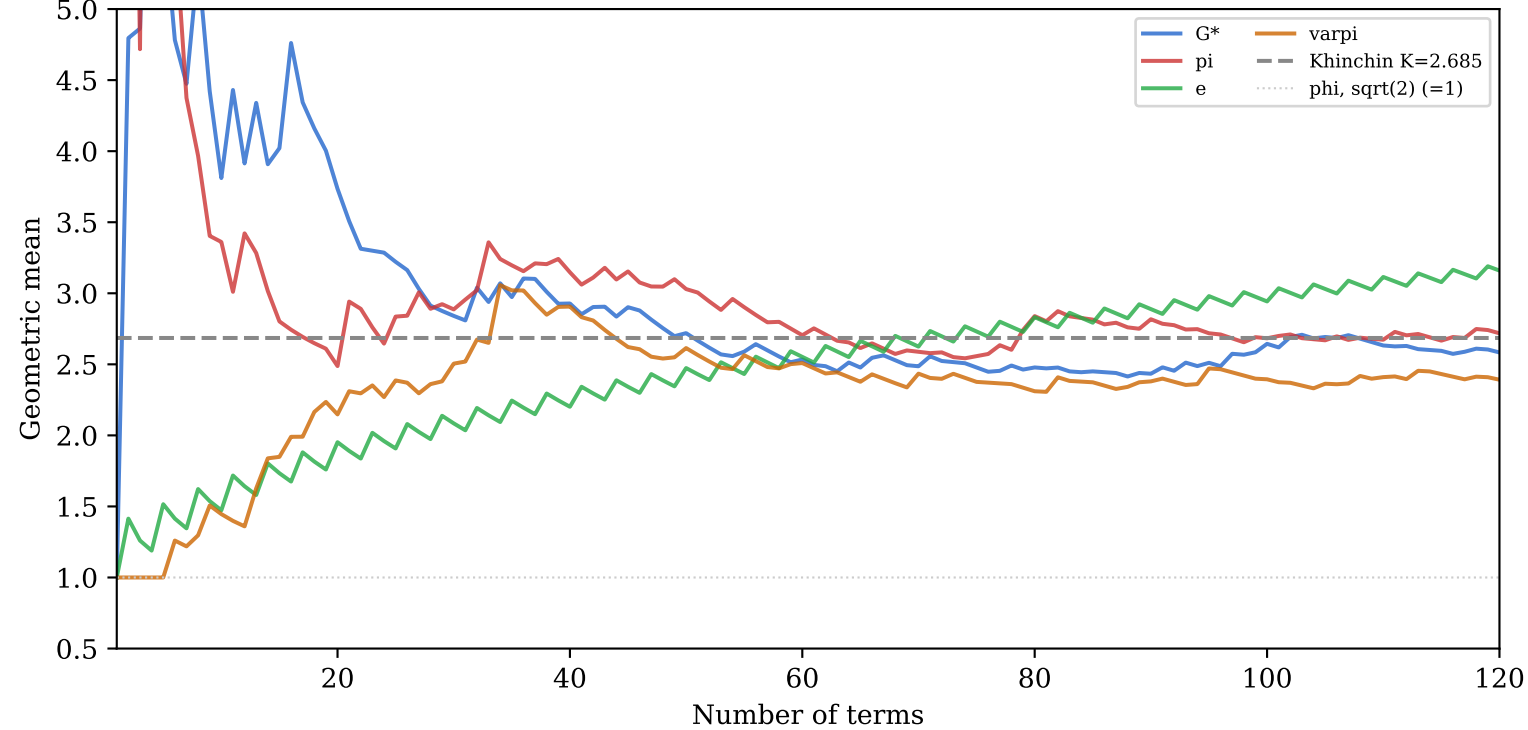
Partial Quotients: First 50 Terms



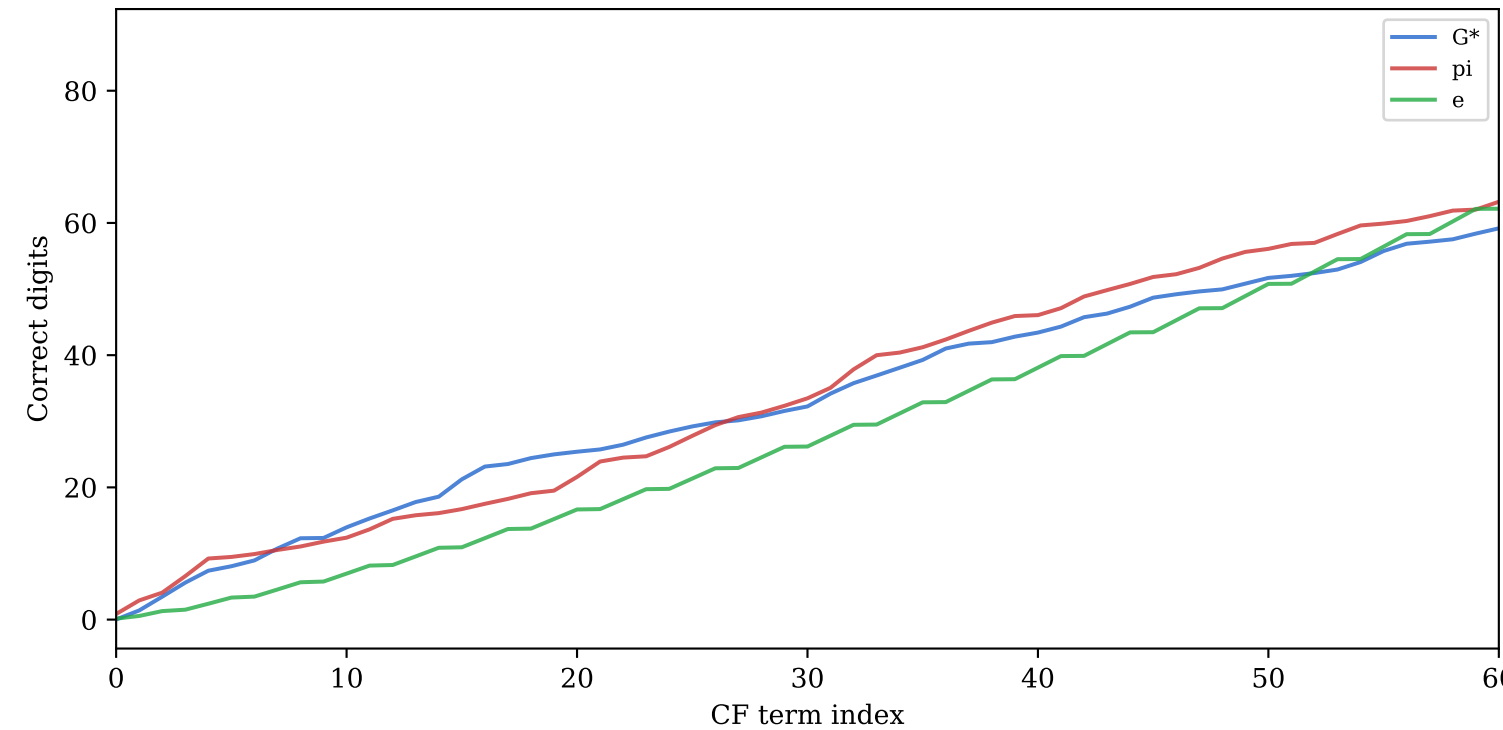
G* Partial Quotients (first 80)



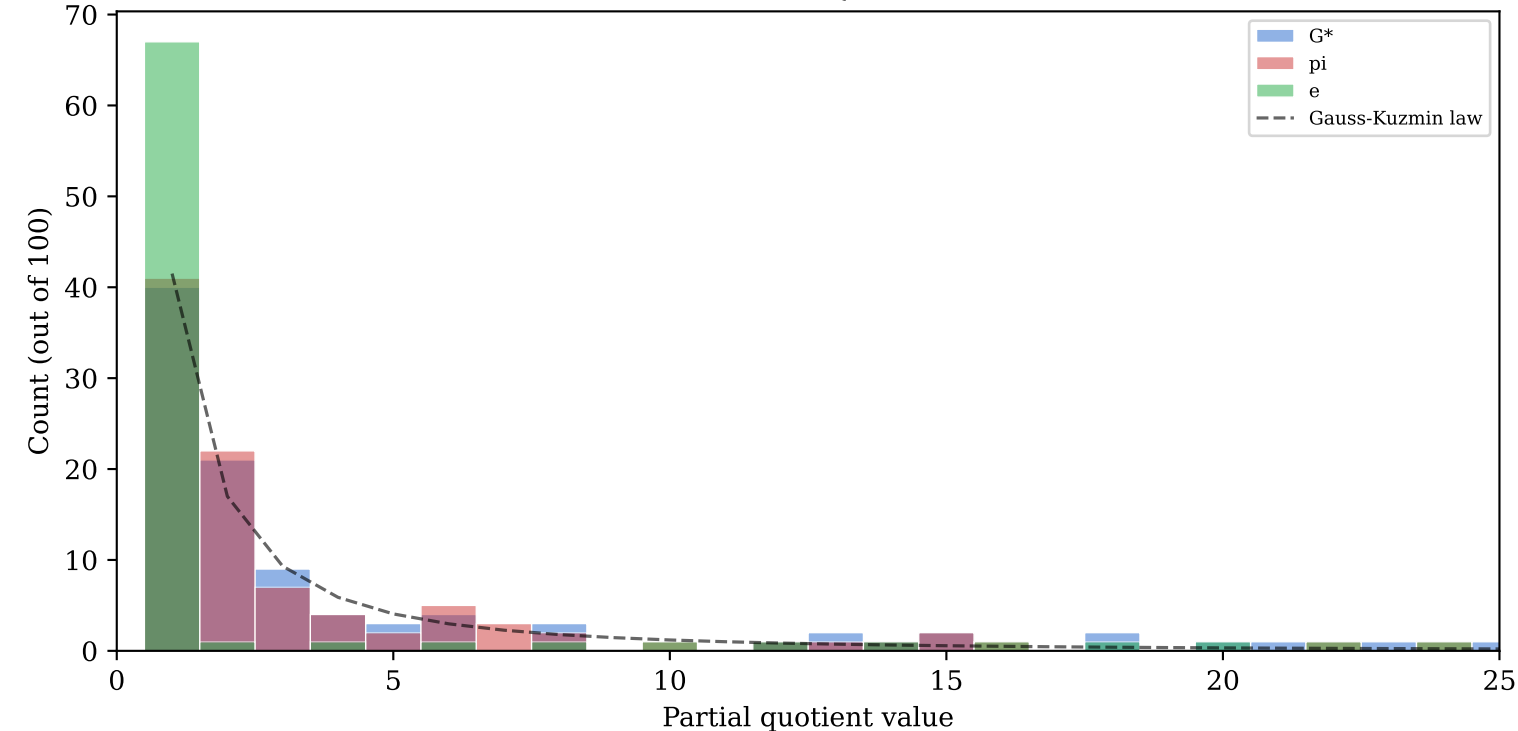
Cumulative Geometric Mean of a_k



Digits of Precision per CF Term



Distribution of Partial Quotients (first 100)



The Verdict

- ϕ , sqrt(2):** **BADLY approximable**
CF = [1; 1, 1, 1, ...] or [1; 2, 2, 2, ...]
Bounded partial quotients. Hardest to approximate by rationals.
- e:** **WELL approximable**
CF = [2; 1, 2, 1, 1, 4, 1, 1, 6, ...]
Periodic large quotients. Easy to approximate.
- π :** **GENERIC (Khinchin-typical)**
CF = [3; 7, 15, 1, 292, 1, 1, 1, ...]
Occasional large quotients. Normal distribution.
- G^* :** **GENERIC (Khinchin-typical)**
CF = [2; 1, 23, 5, 26, 2, 2, 3, ...]
Behaves like a typical real number. No special Diophantine structure.
- varpi:** **GENERIC (Khinchin-typical)**
CF = [2; 1, 1, 1, 1, 1, 4, 1, ...]
Product of two generic constants remains generic.